SPMM Smart Revise

Neurochemistry

Paper A **Syllabic content 3.3**

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1. Synaptic transmission

- The presynaptic neuron synthesises, transports and stores the chemical messenger (neurotransmitter). Synthesis takes place in cell body / soma which contains the essential protein synthesis machinery. From here axonal transport occurs, and the neurotransmitter reaches the synaptic terminal. Before its eventual release, the neurotransmitter is stored within the **synaptic vesicle**. The release takes place through the process of **membrane fusion** and **exocytosis**.
- Upon release, the neurotransmitter occupies receptors present on the surface of the **postsynaptic neuronal membrane.** Some of the neurotransmitter molecules will also act on autoreceptors that are present in the **presynaptic neuronal** membrane. Such **autoreceptor activity** is considered to be crucuial for feedback inhibition of the neurotransmitter synthesis and release.
- Neurotransmitters exhibit **specificity** in receptor interaction. One neurotransmitter can have more than one receptor types, but within a given receptor site only a particular chemical conformation can be accommodated (**lock and key**).
- Receptors have a finite number and thus get **saturated** if there is an over secretion of neurotransmitter.
- Receptor binding is often **competitive**; relative synaptic concentrations of competing molecules decide the eventual degree of receptor activity. Most receptors are bound **reversibly** i.e. following dissociation of the neurotransmitter; the receptor falls back to its physiological status quo. Some molecules can act **irreversibly** producing structural alterations in the protein of receptor complexes.
- After synaptic release and activity, cessation of neurotransmitter action takes place via

    1. Reuptake back to presynaptic neuron via special transporters (e.g. monoamine transporters)
    2. Enzymatic breakdown at the cleft (e.g. via COMT/MAO-A enzyme)
    3. Removed by glia or plasma circulation (e.g. glutamate shuttle)

- Feedback control of a neurotransmitter may exist at various points

    1. Control of presynaptic synthesis
    2. Regulation of release
    3. Reuptake regulation
    4. Autoreceptor mediated presynaptic inhibition
    5. Independent postsynaptic inhibition via a different neuronal network

Neurotransmitters		
Monoamines	**Amino acids**	**Peptides**
Dopamine	GABA	Endorphins
Norepinephrine	Glycine	Cholecystokinin
Epinephrine	Glutamate	Neurotensin
Serotonin		Neuropeptide Y
Acetylcholine		Leptin
Histamine		Ghrelin

2. Classification of receptors

Receptors may be categorized into three categories:

(1) *Ligand-gated* channels (**ionotropic**), in which binding of a chemical messenger alters the probability of opening of transmembrane pores or channels;

(2) Those in which the receptor proteins are coupled to intracellular G proteins as transducing elements (**metabotropic**);

(3) Those termed **ligand-dependent regulators of nuclear transcription** (**nuclear receptors).**

Ionotropic or ion channel receptors result in fast response ($GABA_A$ benzodiazepine); G protein coupling (**metabotropic**) is comparatively a slower process (most antipsychotics, antidepressants).

Ion channel receptors are made up of four or five protein subunits making up a pore like structure. The GABA-A receptor's structure is typical of most ligand-gated (ionotropic) receptors ['doughnut with a hole in the centre' or 'rosette' shaped]. Each protein subunit is a string of amino acids which passes in and out of the cell membrane four times. At the extracellular end of this string is a large N-terminal; this end-chain is thought to mediate GABA-channel interactions. In the middle of the string is a large intracellular loop of amino acids with four sites where phosphorylation occurs.

Inhibitory neurotransmitter action leads to the entry of Cl^- while excitatory action results in the entry of Ca^{2+} or other cations. Ionotropic receptors include $GABA_A$, NMDA, the $5HT_3$ subtype of serotonin receptors.

G-protein-coupled metabotropic receptors are proteins that span the cell membrane seven times (serpentine receptors). G protein-coupled receptors act via cyclase mediated second messenger activation (GTP, ATP, etc.). G_s-proteins are stimulatory; G_i-proteins inhibit the adenylate cyclase. A third variant of G-protein receptors acts via phospholipase C. Metabotropic receptors influence protein synthesis eventually thus producing longer lasting effects. Metabotropic receptors include DA receptors, most 5HT receptors except 5HT-3, NEN and neuropeptides including opioid receptors are G coupled.

Nuclear receptors such as glucocorticoid receptors are part of a superfamily of receptors that have a **cysteine-rich DNA-binding** domain, a ligand-binding domain, and a variable amino terminal region. Upon appropriate ligand binding, a nuclear receptor becomes a **transcription factor** and binds in turn to DNA via **zinc fingers**. Other nuclear receptors include the receptors for progesterone, androgen, and 1,25-dihydroxycholecalciferol (Vitamin D). Many receptors of this family are orphan receptors, for which the ligands are still unidentified.

The glucocorticoid receptor is located mainly in the cytoplasm but migrates to the nucleus as soon as it binds its ligand. In contrast, the estrogen and the triiodothyronine (T_3) receptors are retained in the nucleus and bind hormones directly in the nucleus itself.

3. Dopamine

Source

- tyrosine → l-dopa → dopamine

Rate limiting step

- tyrosine hydroxylase

Breakdown enzymes

- Monoamine oxidase (MAO) & Catechol-o-methyl transferase (COMT).
- MAO-A more selectively metabolizes norepinephrine and serotonin
- MAO-B more selectively metabolizes dopamine.

Breakdown product

- Homovanillic acid

Reuptake

- Dopamine transporter (cocaine inhibits this transported)

Function

- Motivation, novelty seeking, reward circuitry (addictions), arousal and motor movement gating in basal ganglia

Receptors

- 5 types; D_1 to D_5 . All are G protein coupled
- **D_1-like** → D_1 & D_5; increase adenylate cyclase (stimulatory). D1 exclusively postsynaptic; resistant to antagonism. D5 more limbic in distribution; 10 times higher dopamine affinity
- **D_2-like** → $D_{2,3\ \&\ 4}$; decrease adenylate cyclase (inhibitory). D_4 is found primarily in the frontal cortex and clozapine has a high affinity. D_4-selective antagonists do not have antipsychotic efficacy.

Disorders

- Levels low in Parkinson's; high in psychosis especially at mesolimbic area; may be low in anhedonia and negative symptoms in mesocortical area.

4. Noradrenaline

Source

- tyrosine→ l-dopa →dopamine → norepinephrine → epinephrine

Rate limiting step

- tyrosine hydroxylase

Synthetic enzymes

- dopamine-b-hydroxylase modulates norepinephrine production; phenylethanolamine-*N*-methyltransferase modulates conversion of NEN to epinephrine.

Breakdown enzymes

- Monoamine oxidase (MAO – A especially) & Catechol-o-methyl transferase (COMT).

Breakdown product

- 3-methoxy-4-hydroxyphenylglycol (MHPG) & VMA – vanillyl mandelic acid.
- MHPG is the major metabolite in CNS while VMA is major metabolite from peripheral nervous system/endocrine system.

Reuptake

- noradrenaline reuptake channel (tricyclics, reboxetine inhibit this)

Function

- arousal, anxiety, mood regulation, autonomic mediation

Receptors

- **2 major** types; α and β.
- α divided into a_1 and a_2
- α_1 receptors phospholipase C coupled; mostly postsynaptic
- α_2 receptors G_i coupled , mostly presynaptic autoreceptors
- β-receptors G_s coupled; predominate locus ceruleus – may regulate a
- β_1-receptors – high affinity to norepinephrine and β_2-receptors – high affinity to epinephrine.

Disorders

- Levels low in depression and abnormal in panic/anxiety disorders.

5. Serotonin

Source

- tryptophan→5 hydroxy l-tryptophan → serotonin

Rate limiting step

- availability of tryptophan (hence it is possible to conduct tryptophan depletion studies and manipulate 5HT system)

Synthetic enzymes

- tryptophan hydroxylase

Breakdown enzymes

- MAO (preferentially MAO-A)

Breakdown product

- 5-hydroxyindoleacetic acid (5-HIAA)

Reuptake

- Serotonin reuptake channel (tricyclics, SSRIs inhibit this)

Function

- mood, perception of pain, feeding, sleep-wake cycle, motor activity, sexual behaviour, and temperature regulation.

Receptors

- 14 known subtypes of serotonin receptors (5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, 5-HT_{1F}, 5-HT_{2A}, 5-HT_{2B}, 5-HT_{2C}, 5-HT_3, 5-HT_4, 5-HT_{5A}, 5-HT_{5B}, 5-HT_6, and 5-HT_7)
- All except 5-HT_3 are G-protein-coupled receptors; $5HT_3$ predominant in gut; associated with motility.
- 5-HT_{1A} receptors – G_i coupled postsynaptic; antidepressant response; sexual behaviour
- 5-HT_{1B} receptors – G_i coupled presynaptic;
- 5-HT_{1D} receptors – G_i coupled - both presynaptic and postsynaptic.
- 5-HT_2 receptors - phospholipase C coupled; postsynaptic; antagonism leads to antipsychotic response (atypicals) and sedation; LSD causes 5-HT_2 stimulation; down regulation noted after antidepressant treatment / ECT.
- 5-HT_6 may be involved in antidepressant action
- 5-HT7 - regulation of circadian rhythm

Disorders

- low serotonin levels → increased depression, aggression, suicide, and impulsivity; regulate dopamine system – role in psychosis

Receptor	Action
$5HT_{1A}$	Antidepressant (agonist), anxiolytics (partial agonist)
$5HT_{1B}$	Aggression
$5HT_{1D}$	Antimigraine (antagonist)
$5HT_{2A}$	Antipsychotic (antagonist); hallucinogens (agonist / partial agonist); implicated in working memory; also seen in platelets and smooth muscles
$5HT_{2B}$	Stimulation may produce cardiac valvular fibrosis (dexfenfluramine)
$5HT_{2C}$	Anxiogenic and anorexic effect (agonists)
$5HT_{3}$	Antiemetic (antagonist)
$5HT_{6}$	Possible antipsychotic/antidepressant action (antagonism)
$5HT_{7}$	Regulation of circadian rhythm

DOPA decarboxylase (DDC) "is an enzyme implicated in 2 metabolic pathways, synthesizing two important neurotransmitters, dopamine and serotonin (Christenson et al., 1972). Following the hydroxylation of tyrosine to form L-dihydroxyphenylalanine (L-DOPA), catalyzed by tyrosine hydroxylase, DDC decarboxylates L-DOPA to form dopamine. This neurotransmitter is found in different areas of the brain and is particularly abundant in basal ganglia. Dopamine is also produced by DDC in the sympathetic nervous system and is the precursor of the catecholaminergic hormones, noradrenaline and adrenaline in the adrenal medulla". In the nervous system, tryptophan hydroxylase produces 5-OH tryptophan, which is decarboxylated by DDC, giving rise to serotonin. DDC is a homodimeric, pyridoxal phosphate-dependent enzyme. (Excerpt from www.omim.org)

6. Acetylcholine

Source

- choline and acetyl-coenzyme A

Rate limiting step

- availability of choline

Synthetic enzymes

- choline acetyltransferase

Breakdown enzymes

- acetylcholinesterase – rapid metabolism

Breakdown product

- Choline

Reuptake

- no reuptake. Degraded choline is re up-taken and recycled.

Function

- Modulate arousal, learning, memory, rapid eye movement sleep, pain perception, and thirst and parasympathetic mediation.

Receptors

- *Muscarinic receptors* - G-protein-coupled.
- Five subtypes (M_1, M_2, M_3, M_4, and M_5)
- *Nicotinic receptors* - ion channels;
- more in peripheral parasympathetic system;
- Less common than M receptors in CNS – mediates attention.

Disorders

- reduced cholinergic function in Alzheimer's dementia; dopamine balance affected in Parkinson's

7. GABA

Source

- Glutamic acid (glutamate)

Rate limiting step

- glutamic acid decarboxylase (GAD) catalysis

Synthetic enzymes

- glutamic acid decarboxylase (GAD)

Breakdown enzymes

- GABA transaminase

Breakdown product

- Broken down to glutamate, and then eventually to succinic acid

Reuptake

- reuptake into both presynaptic nerve terminals and surrounding glial cells; uptake system is bidirectional and both temperature- and ion-dependent process; (inhibited by tiagabine)

Function

- Mediates anxiety, seizure cessation, and actions of benzodiazepines, barbiturates, and alcohol.

Receptors

- $GABA_A$ and $GABA_B$
- $GABA_A$ – opens chloride channel; inhibitory – leads to hyperpolarization; made of five subunits and at least 14 subunit subtypes
- $GABA_B$ receptor is G-protein-coupled; baclofen is selective agonist

Disorders

- Role in anxiety disorders and alcoholism; may have a role in many other disorders including epilepsy and Huntington's.

8. Glutamate

Source

- 1. from 2-oxoglutarate and aspartate by aspartate aminotransferase,
- 2. from glutamine by glutaminase, or
- 3. from 2-oxoglutarate by ornithine aminotransferase

Regulation

- accumulation of precursors such as glutamine or by end-product inhibition

Synthetic enzymes

- glutaminase

Breakdown enzymes

- Glutamate dehydrogenase, glutamine synthetase

Breakdown product

- Broken down to glutamine or alpha-ketoglutarate

Reuptake

- Largely glial uptake with conversion to glutamine

Function

- Important metabolic role – intermediary in oxidation pathway (malate shuttle), immediate precursor of all GABA in CNS, intermediary in ammonia cycle; NMDA - memory acquisition, developmental plasticity, epilepsy, and ischemic brain injury. NMDA receptor mediates long-term potentiation

Receptors

- *metabotropic* - 8 in total; 3 groups. Group I - $mGluR_1$ & $mGluR_5$ – linked to phospholipase C
- *Ionotropic*: NMDA and non-NMDA
- NMDA - made up of subunits with distinct binding sites for glutamate, glycine, phencyclidine (PCP), magnesium, and zinc.
- Non NMDA – kainate binding or AMPA type.

Disorders

- excitotoxic glutamate toxicity in stroke/schizophrenia/seizures suspected. NMDA antagonists can cause hallucinations – e.g. PCP, ketamine

9. Glycine

- Glycine is the primary inhibitory neurotransmitter in the spinal cord
- It has the simplest structure of all aminoacids
- It is synthesized primarily from serine by serine trans-hydroxymethylase and glycerate dehydrogenase, both of which are rate-limiting steps.
- Glycine acts as a '**mandatory adjunctive neurotransmitter'** for glutamate receptors; the excitatory glycine site on the NMDA receptor is called ***non-strychnine-sensitive*** glycine receptor.
- ***Strychnine-sensitive*** glycine receptor is an inhibitory receptor seen in the spinal cord where glycine acts independently.
- Facilitating glycine transmission can help reduce negative symptoms of schizophrenia. An experimental agent called **bitopertin** is a glycine reuptake inhibitor that has shown some early promise in reducing negative symptoms.

10. Endocannabinoids

- Two endogenous cannabinoid substances - **Anandamide** (a weak ligand) and **2-arachnidonylglycerol** (a strong ligand) are formed from arachidonic acid and ethanolamine.
- The two types of cannabinoid receptors, central (CB_1) and peripheral (CB_2), both bind tetrahydrocannabinol (THC), the active ingredient of marijuana.
- Anandamide lowers intraocular pressure, decreases activity level, and relieves pain.

11. Neurotrophins

These are substances that act as polypeptide growth factors influencing proliferation and differentiation of neurons and glial cells. The best-characterised factors are Nerve growth factor (NGF); brain derived neurotrophic factor (BDNF), neurotrophin 3 and neurotrophin 4.

According to neurotrophin hypothesis neurons compete during development for the limited resource of growth factors in the target region. Those neurons that are highly responsive, e.g. via high affinity binding sites, survive while others undergo programmed cell death. Incorrect targeting of axons may also lead to **apoptosis** (**programmed cell death**).

BDNF may have a role in long-term potentiation (LTP) of memory. In animals, chronic stress leads to down regulation of BDNF. BDNF has been shown to have trophic effects on serotonergic and noradrenergic neurons. SSRIs and other antidepressants including ECT up regulate BDNF. The time course of this up regulation coincides with observed therapeutic actions of antidepressant interventions.

A single nucleotide polymorphism in the BDNF gene on chromosome **11p13** results in an amino-acid substitution of valine (val) with methionine (met) at codon 66 (Val66Met) reducing BDNF activity. BDNF met/met mice demonstrate increased anxiety. Clinical studies in humans have demonstrated that subjects with the Val66Met allele have impaired hippocampal activation and performance. It is controversial if BDNF polymorphism increases the risk of clinical disorders or not.

12. Some clinical implications

Ach & LEWY BODY DEMENTIA

Brain acetylcholine levels are reduced in DLB similar to Alzheimer's. Cortical choline acetyl transferase (ChAT) is reduced to a greater extent (85%) in patients with hallucinations in Lewy body dementia than in those without hallucinations (50%).
This may partially explain the altered sleep-wake patterns seen in DLB and also the response of hallucinations to acetylcholinesterase inhibitors

β ADRENOCEPTOR

Chronic antidepressant treatment induces a reduction in β adrenoreceptor density around 2 weeks after starting antidepressants; this correlates with therapeutic effects.

Unmedicated suicide victims show higher density of β adrenoreceptors.
β blockade can reduce peripheral features of anxiety driven by sympathetic overdrive.

ABERRANT SALIENCE

Kapur proposed that in the normal individual, the role of mesolimbic dopamine is to attach significance or 'salience' to an external stimulus, or an internal thought. This converts a neutral piece of information into an attention grabbing one (Kapur, 2003).

In acute psychosis where hyperdopaminergic state is noted in mesolimbic system, insignificant events and perceptions receive inappropriate salience leading to delusional elaborations.

Antipsychotics are claimed to "dampen the salience" of these abnormal experiences - do not erase the symptoms - but provide the platform for a process of psychological resolution.

5HT & DEPRESSION

An increased density of $5HT_2$ binding sites has been shown in post mortem studies of depressed / suicidal patients. The increase in $5HT_{2A}$ receptors is most prominent in dorsolateral prefrontal cortex and in platelets of medication naïve patients. A reduction in $5HT_{1A}$ receptors has also been noted in cortex

Long-term antidepressant treatment has been shown to reduce $5HT_2$ receptors and increase $5HT_{1A}$ function. But these changes may not be causative of antidepressant action as they predate any clinical response to antidepressant therapy

Most directly acting $5HT_{1A}$ agonists have poor antidepressant activity.

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug related information using external sources; no part of these notes should be used as prescribing information.

Notes prepared using excerpts from:

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